mhs-poisson-z: a pressure-only magnetohydrostatic correction on a geometric-height grid

Physics and method of `run.mode = "mhs-poisson-z"` in HSEMHS. A self-contained description.

Abstract

Starting from an atmosphere whose columns are each in hydrostatic equilibrium, this mode adjusts the gas pressure in three dimensions so that pressure, gravity and the Lorentz force of a given magnetic field balance each other as well as possible. Temperature and magnetic field stay fixed; only the pressure changes. Because a fixed field generally cannot hold up every horizontal density contrast, an exact equilibrium does not exist, and the pressure is found as the minimiser of a weighted sum of squared relative force residuals. We derive the force balance, show why its pressure part is a Poisson problem, explain what a pressure change can and cannot achieve, define the residuals and their weights, and describe the Gauss–Newton iteration, the Fourier-mode preconditioner, the step control, the stopping rule, the treatment of the low-β upper atmosphere and the computation of the optical depth. All assumptions, including how the pressure level is fixed, are listed.

1. Introduction

A hydrostatic atmosphere in which every vertical column carries its own weight separately has two shortcomings. First, neighbouring columns of different temperature have different pressures at the same height, so horizontal pressure forces remain unbalanced. Second, the magnetic field is ignored. This mode keeps the temperature T and the magnetic field $\mathbf{B}$ exactly as given and asks: *which pressure field balances, as well as possible, the pressure force, gravity and the magnetic force in all three directions?* The idea comes from the FIRTEZ-dz inversion code (Borrero et al.), where the atmosphere changes in every inversion cycle and a pressure-only correction restores the force balance far faster than a correction of the field would.

Input: T and $\mathbf{B}$ on a three-dimensional geometric-height grid that is periodic in x and y; the gravity g; a reference pressure P_{ref} for the starting state; the height z_{cut} below which the force balance is solved. **Output:** P, ρ and τ_{500} on the grid, the height of $\tau_{500} = 1$ in every column, and the starting (column-hydrostatic) pressure for comparison.

2. Notation

Symbol	Meaning	Unit	Role
$x,\ y$	horizontal coordinates, periodic; x_i, $i = 1 \dots N_x$, y_j, $j = 1 \dots N_y$, spacing Δx, Δy	m	given
z	height (upward); levels z_k, $k = 1 \dots N_z$, top first	m	given
$T,\ X$	temperature; $X = \ln T$	K	given, fixed
$\mathbf{B}$	magnetic field at the grid points	T	given, fixed
$\mathbf{J}$	current density $\nabla \times \mathbf{B}/\mu_0$	$\mathrm{A\,m^{-2}}$	fixed
$\mathbf{L}$	Lorentz force per unit volume $\mathbf{J} \times \mathbf{B}$	$\mathrm{N\,m^{-3}}$	fixed
$P,\ u$	gas pressure; $u = \ln(P/1\,\mathrm{Pa})$	Pa, –	unknown
P_{ref}	pressure at the bottom of every column in the starting state	Pa	parameter
$\rho,\ q$	density $\rho(T, P)$; $q = \rho/P$	$\mathrm{kg\,m^{-3}}$, $\mathrm{s^2\,m^{-2}}$	derived
$H,\ \bar{H}_k$	scale height $P/(\rho g)$; its median over layer k in the starting state	m	derived, fixed
$\mathbf{F}$	net force per unit volume	$\mathrm{N\,m^{-3}}$	residual
$\mathbf{R}$	relative residual $\mathbf{F}/P$	$\mathrm{m^{-1}}$	residual
Φ	objective: weighted sum of squared residuals	m	minimised
β	plasma beta $2\mu_0 P/B^2$	–	derived
$z_{\mathrm{cut}},\ k_c$	top of the solved region (default 1 Mm); its level index	m, –	parameter
n	number of solved levels, $n = N_z - k_c + 1$	–	derived
h_k	level spacing $z_k - z_{k+1}$ in the solved region	m	grid
w_k	thickness of the cell around level k	m	grid
δu	change of u in one Gauss–Newton step	–	step
J, W	Jacobian matrix $\partial\mathbf{R}/\partial u$; diagonal matrix of the weights		derived
g	gravitational acceleration, from the input file (274)	$\mathrm{m\,s^{-2}}$	parameter
μ_0	vacuum permeability $4\pi \times 10^{-7}$	$\mathrm{T\,m\,A^{-1}}$	constant

3. The force balance

3.1 The three forces

A small volume of gas feels three forces per unit volume:

1. the **pressure force** $-\nabla P$, pushing gas from high to low pressure;
2. **gravity** $-\rho g\,\hat{\mathbf{z}}$;
3. the **Lorentz force** $\mathbf{L} = \mathbf{J} \times \mathbf{B}$ of the magnetic field on the electric currents in the ionised gas. In a slowly changing plasma the currents follow from the field by Ampère's law, $\mathbf{J} = \nabla \times \mathbf{B}/\mu_0$.

With a vector identity the Lorentz force splits into two parts with a clear meaning:

$$\mathbf{J} \times \mathbf{B} = \frac{(\nabla \times \mathbf{B}) \times \mathbf{B}}{\mu_0} = \underbrace{-\nabla\left(\frac{B^2}{2\mu_0}\right)}_{\text{magnetic pressure}} + \underbrace{\frac{(\mathbf{B}\cdot\nabla)\mathbf{B}}{\mu_0}}_{\text{magnetic tension}} . \tag{1}$$

Field lines push sideways on each other like a gas with pressure $B^2/2\mu_0$, and curved field lines pull like stretched rubber bands.

3.2 Magnetohydrostatic equilibrium in three components

The gas is static when the three forces cancel:

$$\boxed{\mathbf{F} \equiv -\nabla P + \mathbf{J} \times \mathbf{B} - \rho\, g\, \hat{\mathbf{z}} = 0} \tag{2}$$

or, component by component,

$$-\frac{\partial P}{\partial x} + L_x = 0, \qquad -\frac{\partial P}{\partial y} + L_y = 0, \qquad -\frac{\partial P}{\partial z} - \rho g + L_z = 0. \tag{3}$$

In this mode $\mathbf{L}$ is a *given* force field, computed once from the fixed $\mathbf{B}$, and ρ depends on P through the equation of state. The question is therefore which single scalar field $P(x, y, z)$ balances gravity plus a given vector force field.

3.3 Why it is called a Poisson step

Take the divergence of Eq. (2), using $\nabla \cdot \nabla P = \nabla^2 P$:

$$\nabla^2 P = \nabla \cdot (\mathbf{J} \times \mathbf{B}) - g\,\frac{\partial \rho}{\partial z}. \tag{4}$$

This is a **Poisson equation** for P, the same type of equation as that for the electric potential of a charge distribution, with a right-hand side that itself depends on P through ρ. FIRTEZ solves essentially this equation, iterating on the density. Poisson problems in a horizontally periodic box can be solved very fast with Fourier transforms (Sect. 10.4). The method used here does not solve Eq. (4) literally but minimises the residual of Eq. (2) (Sect. 6); each of its linear steps turns out to be a Poisson-type equation.

4. The equation of state and the opacity

Every force balance in this document involves the density ρ, and every optical depth involves the extinction coefficient χ_{500}. Neither is an unknown of its own: at given temperature and pressure both are fixed by the microphysics of the gas. This section explains how.

4.1 The ideal gas

In the solar photosphere and chromosphere the particles interact only weakly, so the gas obeys the ideal-gas law. If n is the number of free particles (atoms, ions, molecules and electrons) per cubic metre,

$$P = n\, k_B T. \tag{5}$$

If the average mass of a free particle is $\bar{\mu}\, m_{\mathrm{u}}$ (with m_{u} the atomic mass unit and $\bar{\mu}$ the **mean molecular weight**), the mass density is $\rho = n\,\bar{\mu}\, m_{\mathrm{u}}$, and therefore

$$\rho = \frac{\bar{\mu}\, m_{\mathrm{u}}}{k_B}\,\frac{P}{T}. \tag{6}$$

At fixed $\bar{\mu}$ and T the density is proportional to the pressure.

4.2 Why $\bar{\mu}$ depends on T and P: ionisation and molecules

For neutral solar gas (about 91 % hydrogen and 9 % helium atoms by number, plus a small fraction of heavier elements) $\bar{\mu} \approx 1.3$. When a hydrogen atom is ionised it becomes two free particles, a proton and an electron, of

almost the same total mass; in fully ionised gas $\bar{\mu} \approx 0.6$. In the coolest layers hydrogen partly forms H_2 molecules, which raises $\bar{\mu}$. The degree of ionisation of an element in stage i is given by the **Saha equation**

$$\frac{n_{i+1}\, n_e}{n_i} = \frac{2U_{i+1}(T)}{U_i(T)} \left(\frac{2\pi m_e k_B T}{h^2} \right)^{3/2} e^{-E_i/k_B T}, \tag{7}$$

with E_i the ionisation energy (13.6 eV for hydrogen), U_i the partition functions, m_e the electron mass and h Planck's constant. Two properties matter for us. (i) Because of the exponential, ionisation switches on over a narrow range of temperature. (ii) The electron density n_e appears on the left: at higher density there are more electrons to recombine with, and a higher temperature is needed for the same ionisation. Hence $\bar{\mu} = \bar{\mu}(T, P)$, and Eq. (6) is no longer a simple proportionality. The Saha equations of all elements are coupled through the common n_e and through charge conservation (n_e equals the total charge of the ions), and the molecular equilibria (H_2, H_2^+, H^-) are solved together with them. This assumes **local thermodynamic equilibrium** (LTE): the level populations and ionisation stages are those of a gas in equilibrium at the local T.

4.3 The 500 nm extinction

Light of wavelength 500 nm travelling a distance ds loses the fraction $\chi_{500}\, ds$ of its intensity, by absorption and by scattering. In the solar photosphere χ_{500} is dominated by the negative hydrogen ion H^- (a hydrogen atom with a second, weakly bound electron, binding energy 0.75 eV), whose abundance is proportional to the electron density. The electrons come mostly from easily ionised metals (Na, Mg, Si, Ca, Fe, ...), so χ_{500} grows steeply with temperature and with density. The code adds the continuum contributions of H (bound–free and free–free), H^-, H_2^+, He, He^-, the metals C, N, Mg, Si, Ca and Fe, Rayleigh scattering on H, He and H_2, and Thomson scattering on free electrons, each counted once. Spectral lines are not included: χ_{500} is a continuum extinction.

4.4 The table and its interpolation

Solving the coupled Saha, molecular and charge equations at every grid point at every step would be far too slow. They are therefore solved once, with Wittmann's equation-of-state code for solar composition (with a dedicated hot branch above a switch temperature $T_{\rm sw}$, where the original solver fails in the hot, low-pressure corner), on a grid of $X = \ln T$ and $Y = \ln P$, and stored in a table. For every node the table holds the three functions

$$\ln \rho(X, Y), \qquad \ln n_e(X, Y), \qquad \ln \chi_{500}(X, Y),$$

together with their derivatives ∂_X, ∂_Y and $\partial_X \partial_Y$. Between the nodes each function is evaluated by **bicubic Hermite interpolation**: a cubic polynomial in each direction that matches the stored values and derivatives at the four corners of the cell. The result is continuous with continuous first derivatives, which is important for the Newton-type solvers of the magnetohydrostatic modes, and it provides the derivative $\partial \ln \rho / \partial \ln P|_T$ that they need. The table used by all modes is `cache/eos_table_fullz.h5`. Its range of T and P (its *envelope*) covers the input atmospheres; a point outside the envelope is reported as an equation-of-state failure, never extrapolated.

Two derived quantities appear repeatedly:

$$q \equiv \frac{\rho}{P} \quad [\mathrm{s^2\, m^{-2}}], \qquad H \equiv \frac{P}{\rho\, g} = \frac{1}{g\, q} = \frac{k_B T}{\bar{\mu}\, m_{\rm u}\, g} \quad [\mathrm{m}]. \tag{8}$$

H is the **pressure scale height**. For $T = 6000$ K and $\bar{\mu} = 1.3$, $H \approx 140$ km.

5. What a pressure correction can do, and what it cannot

5.1 Gradient and non-gradient forces

A force field in the box can be split (the **Helmholtz decomposition**) into

$$\mathbf{F} = \underbrace{-\nabla\phi}_{\text{gradient part}} + \underbrace{\mathbf{S}}_{\text{non-gradient part}}, \qquad \int \mathbf{S} \cdot \nabla\psi \, dV = 0 \text{ for every function } \psi. \tag{9}$$

The gradient part is the slope of a scalar potential ϕ, like a pressure force, and has no curl. The non-gradient part carries the whole curl of $\mathbf{F}$, and the orthogonality condition means, after integration by parts, that it has no divergence. The two parts are orthogonal, so their sizes add like the sides of a right triangle, $\int |\mathbf{F}|^2 dV = \int |\nabla\phi|^2 dV + \int |\mathbf{S}|^2 dV$. The split is unique once boundary conditions are fixed: the box is periodic in x and y but has a top and a bottom, where conditions (and an additive constant in ϕ) have to be specified.

A change of pressure δP changes the force by $-\nabla\delta P$, a pure gradient. If the density did not change, a pressure correction could remove the gradient part ($\delta P = -\phi$) and could never touch $\mathbf{S}$. This picture explains what the method can achieve; the solver itself does not compute such a projection. It minimises a weighted sum of squares of the relative residual (Sect. 6), a non-linear problem, so it neither removes the gradient part exactly nor leaves $\mathbf{S}$ untouched.

5.2 Which forces are not gradients?

Take the curl of Eq. (2). The curl of a gradient is zero, and $\nabla \times (\rho g \hat{\mathbf{z}}) = g\,(\partial_y \rho,\, -\partial_x \rho,\, 0)$, so

$$\nabla \times \mathbf{F} = \nabla \times (\mathbf{J} \times \mathbf{B}) - g \left(\frac{\partial \rho}{\partial y},\, -\frac{\partial \rho}{\partial x},\, 0 \right). \tag{10}$$

Term by term: the **pressure** force is always a gradient. **Gravity** is a gradient only if ρ is uniform on each horizontal plane; a heavy column next to a light one at the same height cannot be static (**buoyancy**). The **Lorentz force** has a gradient part (the magnetic pressure) and in general a non-gradient part (the tension). Static equilibrium therefore requires the **curl condition**

$$\nabla \times (\mathbf{J} \times \mathbf{B}) = g \left(\frac{\partial \rho}{\partial y},\, -\frac{\partial \rho}{\partial x},\, 0 \right) : \tag{11}$$

only the tension of the field can hold up horizontal density differences. Without a field this forces $\partial_x \rho = \partial_y \rho = 0$. With a given $\mathbf{B}$, Eq. (11) holds only if the field happens to have exactly the right tension everywhere. There is no reason to expect this of a field taken from a moving simulation or an inversion, so in general *no pressure makes the atmosphere exactly static*, and the mode looks for the best compromise.

5.3 The density loophole

The density is not fixed: at fixed T a pressure change changes it,

$$\delta\rho = \rho \left. \frac{\partial \ln \rho}{\partial \ln P} \right|_T \frac{\delta P}{P}, \qquad \text{so} \qquad \delta(\nabla \times \mathbf{F}) = -g\, \nabla \times (\delta\rho\, \hat{\mathbf{z}}). \tag{12}$$

By making the pressure fall faster with height under a dense column, the correction lowers that column's density at a given height and reduces the horizontal density contrast, and with it part of $\mathbf{S}$. What it can never remove is the part caused by temperature differences at fixed pressure or by the tension of $\mathbf{B}$: that requires changing $\mathbf{B}$ or T.

6. The quantity minimised

6.1 Relative residual

Divide the force by the pressure and use $u = \ln P$ ($\nabla P/P = \nabla u$, $\rho g/P = g\,q = 1/H$):

$$\boxed{\mathbf{R} \equiv \frac{\mathbf{F}}{P} = -\nabla u - g\, q(T,u)\, \hat{\mathbf{z}} + \mathbf{L}\, e^{-u}} \tag{13}$$

1. $P = e^u$ stays positive across the many decades of pressure in the box.
2. $\mathbf{R}$ is a *relative* imbalance. Minimising $|\mathbf{F}|$ itself would reward making the pressure small everywhere (all forces scale with P); dividing by P removes that false incentive.
3. Without the Lorentz force, $R_z = 0$ is $du/dz = -1/H$, ordinary hydrostatic equilibrium.
4. $|\mathbf{R}|\, H \approx |\mathbf{F}|/(\rho g)$: multiplied by the scale height, the residual is the unbalanced force in units of the weight of the gas. A value of 1 means an unbalanced force as large as gravity.

6.2 More equations than unknowns: least squares

At every point the force balance gives three equations, but there is only one unknown, the pressure. With more equations than unknowns and, by Sect. 5, no exact solution, we look for the **best compromise**: the pressure that minimises the weighted sum of squared residuals, just as fitting a straight line through many points minimises the squared distances. The objective is a discrete volume integral,

$$\boxed{\Phi(u) = \sum_{k,j,i} w_k\, \bar{H}_k^2 \left[(R_x)_{ijk}^2 + (R_y)_{ijk}^2 \right] + \sum_{k,j,i} h_k\, \bar{H}_{k+1/2}^2\, (R_z)_{ij,k+1/2}^2} \tag{14}$$

(the discrete residuals are defined in Sect. 9). w_k and h_k are the thicknesses of the cells around the levels and between them, so Φ approximates $\int |\mathbf{R}|^2 \bar{H}^2\, dz$ summed over the columns. $\bar{H}_k$ is the median of H over layer k in the starting state and stays fixed during the solve; $\bar{H}_{k+1/2} = \sqrt{\bar{H}_k \bar{H}_{k+1}}$. The factor $\bar{H}^2$ makes each term a force in units of the local weight, so that the deep, dense layers, whose forces are orders of magnitude larger, do not swamp the upper ones. The reported measure of the imbalance is $\mathrm{RMS}\,|\mathbf{R}|\bar{H} = \sqrt{\Phi/W}$, with W the sum of all weights (w_k twice, for x and y, and h_k) over all columns.

6.3 The pressure level: the reference pressure

Force balance hardly fixes the overall level of the pressure. For an ideal gas of fixed ionisation, multiplying P everywhere by a constant leaves ∇u and q unchanged, so only the Lorentz term $\mathbf{L}e^{-u}$ and the weak pressure dependence of the ionisation would notice. The level is therefore fixed separately, in two stages:

1. The **starting state** is the column-hydrostatic atmosphere with the same pressure P_{ref} at the bottom of every column (for the production input, the median of the simulation's own pressure over the bottom plane). It is computed by integrating $du/dz = -g\,q$ upward from the bottom in every column (classical Runge–Kutta with substep doubling to a tolerance of 10^{-6} in u). The bottom is used because there, deep below the visible surface, the true pressure varies least across a horizontal plane.
2. During the solve each step is penalised for changing the **mean of u over the bottom plane** (Sect. 10.2). The pattern of u on the bottom plane is free to change, so the horizontal forces there can be balanced; only its mean is held near the starting value $\ln P_{\mathrm{ref}}$. This is a penalty, not an exact constraint, and the limit on the size of each step can shift it slightly.

7. The low-β problem and the cut at z_{cut}

The plasma beta $\beta = 2\mu_0 P/B^2$ compares gas and magnetic pressure. Deep down $\beta \gg 1$: the gas dominates. Higher up $\beta \ll 1$: the Lorentz force dwarfs every gas force, and a static state needs a nearly **force-free** field, $\mathbf{J} \times \mathbf{B} \approx 0$ (currents along the field lines). A field from a dynamic simulation or an inversion is not force-free. With $\mathbf{B}$ fixed, the only way to balance a large Lorentz force is a large pressure gradient, i.e. a large pressure; and because $\partial P/\partial z$ links each layer to the one below, raising the pressure high up raises it in the photosphere too, which would destroy the result where it matters most.

The force balance is therefore solved only on the levels with $z_k \le z_{\mathrm{cut}}$ (default 1 Mm), where the gas still dominates. Above z_{cut} each column is integrated in plain hydrostatic equilibrium, $du/dz = -g\,q$, upward from the solved pressure at the first level below the cut (Sect. 11). FIRTEZ makes the same physical choice more gradually (its magnetic term decays exponentially above a chosen height). z_{cut} is a physical parameter: it marks where we stop trusting the given field to be in equilibrium.

8. Assumptions

A1. Static gas: velocities neglected.
A2. Fixed temperature and field. T and $\mathbf{B}$ are not changed; hence $\mathbf{J}$ and $\mathbf{L}$ are fixed, computed once.
A3. Best compromise, not equilibrium: the solution minimises Eq. (14); an exact equilibrium generally does not exist (Sect. 5).
A4. Weights $\bar{H}_k$ from the starting state, fixed during the solve.
A5. Pressure level: equal P_{ref} at the bottom of every column in the starting state; afterwards the mean of $\ln P$ over the bottom plane held by a per-step penalty (Sect. 6.3).
A6. Horizontally periodic box; no condition is imposed at the top and bottom of the solved region other than those implied by the residuals there (the natural boundary conditions of the least-squares problem).
A7. Hydrostatic above z_{cut}: no Lorentz force and no horizontal balance there.
A8. LTE ideal gas with ionisation and molecules from the table; **constant vertical gravity**, plane-parallel geometry.
A9. Discretisation: field and pressure at the same grid points; second-order differences.

9. The discrete problem

9.1 Grid of the solved region

The solved region consists of the levels $k_c \dots N_z$ (k_c the first level with $z_k \le z_{\mathrm{cut}}$), renumbered $k = 1 \dots n$ from the top. Every unknown u_{ijk} sits at a grid point. Between levels k and $k+1$ the spacing is $h_k = z_k - z_{k+1}$ and the face lies at $z_{k+1/2} = \frac{1}{2}(z_k + z_{k+1})$; the outer faces lie half a spacing outside the end levels. The cell thickness is $w_k = z_{k-1/2} - z_{k+1/2}$. Altogether there are nN_xN_y unknowns and $(3n-1)N_xN_y$ residuals.

9.2 The Lorentz force

The current is computed on the full grid with second-order differences, centred and periodic in x and y,

$$(\partial_x f)_{ijk} = \frac{f_{i+1,jk} - f_{i-1,jk}}{2\Delta x}, \qquad (\partial_y f)_{ijk} = \frac{f_{i,j+1,k} - f_{i,j-1,k}}{2\Delta y},$$

and in z with the second-order three-point formula for unequal spacing (one-sided at the top and bottom levels). Then

$$\mathbf{J} = \frac{1}{\mu_0}\big(\partial_y B_z - \partial_z B_y,\ \partial_z B_x - \partial_x B_z,\ \partial_x B_y - \partial_y B_x\big), \qquad \mathbf{L} = \big(J_y B_z - J_z B_y,\ J_z B_x - J_x B_z,\ J_x B_y - J_y B_x\big). \quad (15)$$

9.3 The residuals

The horizontal residuals live at the grid points,

$$(R_x)_{ijk} = -\frac{u_{i+1,jk} - u_{i-1,jk}}{2\Delta x} + (L_x)_{ijk}\, e^{-u_{ijk}}, \qquad (R_y)_{ijk} = -\frac{u_{i,j+1,k} - u_{i,j-1,k}}{2\Delta y} + (L_y)_{ijk}\, e^{-u_{ijk}}, \tag{16}$$

and the vertical one on the interior faces $k+1/2$, $k = 1 \dots n-1$:

$$(R_z)_{ij,k+1/2} = -\frac{u_{ijk} - u_{ij,k+1}}{h_k} - \frac{g}{2}\big(q_{ijk} + q_{ij,k+1}\big) + \frac{1}{2}\big((L_z)_{ijk} e^{-u_{ijk}} + (L_z)_{ij,k+1} e^{-u_{ij,k+1}}\big), \tag{17}$$

with $q_{ijk} = \rho(T_{ijk}, e^{u_{ijk}})\, e^{-u_{ijk}}$. Placing R_z between the levels makes the vertical pressure difference a compact two-point difference; the density and the Lorentz force are averaged to the face.

10. The Gauss–Newton iteration

10.1 Linearisation

$\mathbf{R}$ depends on u linearly through $-\nabla u$, but non-linearly through $q(T,u)$ and the factor e^{-u}. Newton's method for one equation $f(u) = 0$ replaces f by its tangent and solves $f(u) + f'(u)\,\delta u = 0$. For a least-squares problem the analogue is **Gauss–Newton**: replace the residual vector by its linear approximation in the step,

$$\mathbf{R}(u + \delta u) \approx \mathbf{R}(u) + \mathsf{J}\,\delta u, \qquad \mathsf{J} = \frac{\partial \mathbf{R}}{\partial u}, \tag{18}$$

and minimise the resulting quadratic $\Phi(u+\delta u) \approx (\mathbf{R} + \mathsf{J}\delta u)^{\top}\mathsf{W}(\mathbf{R} + \mathsf{J}\delta u)$. The derivatives are

$$\delta(R_x)_{ijk} = -\frac{\delta u_{i+1,jk} - \delta u_{i-1,jk}}{2\Delta x} - (L_x)_{ijk} e^{-u_{ijk}}\, \delta u_{ijk} \quad \text{(and likewise for } y\text{)}, \tag{19}$$

$$\delta(R_z)_{ij,k+1/2} = -\big(a_k\, \delta u_{ijk} + b_k\, \delta u_{ij,k+1}\big), \tag{20}$$

$$a_k = \frac{1}{h_k} + \frac{1}{2}\Big(g\, q'_{ijk} + (L_z)_{ijk} e^{-u_{ijk}}\Big), \qquad b_k = -\frac{1}{h_k} + \frac{1}{2}\Big(g\, q'_{ij,k+1} + (L_z)_{ij,k+1} e^{-u_{ij,k+1}}\Big), \tag{21}$$

with $q' = \partial q/\partial u = q\,\big(\partial \ln\rho/\partial \ln P|_T - 1\big)$, taken from the derivative of the table interpolation. (a_k and b_k also depend on i, j; the indices are dropped for brevity.)

10.2 The normal equations

Setting the derivative of the quadratic with respect to δu to zero gives the **normal equations**

$$\boxed{\big(\mathsf{J}^{\top}\mathsf{W}\mathsf{J} + \mathsf{E} + \mathsf{A}\big)\,\delta u = -\mathsf{J}^{\top}\mathsf{W}\,\mathbf{R}} \tag{22}$$

with two additional terms:

- E, a small diagonal damping, $E_k = 10^{-2}\,\kappa_1^2\, w_k \bar{H}_k^2$ with $\kappa_1 = 2\pi/(N_x\Delta x)$ the smallest horizontal wavenumber of the box: one per cent of the horizontal stiffness of the longest wave. It keeps the system well posed for components that the residuals barely see.
- A, the **anchor** penalty of Sect. 6.3: the term $\frac{1}{2}A\big(\sum_{\text{bottom}} \delta u\big)^2$ is added to the quadratic, so $\mathsf{A} = A\,\mathbf{c}\mathbf{c}^{\top}$ with $\mathbf{c}$ the indicator vector of the bottom-plane points and $A = w_{\mathrm{a}}\,\bar{H}^2_{n-1/2}/(h_{n-1}N_xN_y)$ (w_{a} = `anchor_w`, default 1), comparable to the vertical stiffness of the lowest face.

The right-hand side $-\mathsf{J}^{\top}\mathsf{W}\mathbf{R}$ is minus half the gradient of Φ. Since J contains the difference operators $-\nabla$, the leading part of $\mathsf{J}^{\top}\mathsf{W}\mathsf{J}$ is $-\nabla\cdot\big(w\bar{H}^2\nabla\,\cdot\big)$, a weighted Laplacian: *each step is a Poisson equation for* δu, plus local terms from the equation of state and the Lorentz force. Gauss–Newton neglects the second derivatives of the residuals. That is accurate where the residuals are small, but when a sizeable residual remains at the minimum, as here, the convergence is only linear: Φ decreases by a roughly constant factor per step rather than ever faster.

10.3 Solving the normal equations: preconditioned conjugate gradients

Equation (22) is a symmetric, positive-definite linear system with as many unknowns as solved grid points (tens of millions). It is never written out as a matrix. The **conjugate-gradient** (CG) method needs only products of the matrix with a vector, and each product applies J and $\mathsf{J}^{\top}$ with the stencils above. CG starts from $\delta u = 0$ and builds the solution from successive search directions, each conjugate to the previous ones. It converges in few iterations if a good **preconditioner** $\mathsf{M} \approx \mathsf{J}^{\top}\mathsf{W}\mathsf{J} + \mathsf{E} + \mathsf{A}$ with a cheap inverse is available: CG is then applied in effect to $\mathsf{M}^{-1}(\mathsf{J}^{\top}\mathsf{W}\mathsf{J} + \mathsf{E} + \mathsf{A})$, which is close to the identity. CG stops when the residual of the linear system has fallen to 10^{-3} of its starting value (`cg_tol`) or after 80 iterations (`cg_maxit`). An inexact solution is acceptable, because the step is in any case only the minimiser of a linear model.

10.4 The preconditioner: Fourier modes in x and y

In a horizontally periodic box any function can be written as a sum of waves, $\delta u_{ijk} = \sum_{k_x,k_y} \hat{u}_k(k_x,k_y)\, e^{\,i(k_x x_i + k_y y_j)}$. A centred difference acts on a wave as a multiplication, $\partial_x \to i\sin(k_x\Delta x)/\Delta x$, so the horizontal part of the operator becomes a multiplication by $\sigma^2 = \sin^2(k_x\Delta x)/\Delta x^2 + \sin^2(k_y\Delta y)/\Delta y^2$. If all coefficients were the same throughout each horizontal layer, every wave (k_x,k_y) would obey its own one-dimensional equation in z. The preconditioner makes exactly this approximation: the coefficients a_k, b_k are replaced by their medians over each layer, and the local Lorentz terms of the horizontal residuals are dropped. For each wave the vertical system is then tridiagonal,

$$\begin{aligned} \big(C_k + w_k \bar{H}_k^2\,\sigma^2 + E_k\big)\hat{u}_k + D_{k-1}\hat{u}_{k-1} + D_k\hat{u}_{k+1} &= \hat{r}_k, \\ C_k = h_{k-1}\bar{H}_{k-1/2}^2\, b_{k-1}^2 + h_k\bar{H}_{k+1/2}^2\, a_k^2, \qquad D_k &= h_k\bar{H}_{k+1/2}^2\, a_k b_k, \end{aligned} \tag{23}$$

and is solved in a number of operations proportional to n (Thomas algorithm). The horizontal mean wave ($k_x = k_y = 0$) also carries the anchor term, a rank-one matrix, which is included exactly with the Sherman–Morrison formula on top of the tridiagonal solve. Applying M^{-1} costs one forward two-dimensional Fourier transform per layer, one tridiagonal solve per wave and one inverse transform. The medians are recomputed at every Gauss–Newton step.

10.5 Step control

The solution δu of Eq. (22) is the step the linear model recommends. Two safeguards protect against the non-linearity:

1. **Cap:** every component is clamped to $|\delta u_{ijk}| \le 2$ (the pressure may change by at most a factor $e^2 \approx 7.4$ per point and step).
2. **Backtracking:** try $u + \alpha\,\delta u$ with $\alpha = 1, \tfrac{1}{2}, \tfrac{1}{4}, \ldots$ (at most 12 trials) and accept the first trial that lowers Φ and stays inside the equation-of-state table.

If no trial lowers Φ, the iteration stops.

10.6 Stopping rule

After each accepted step the relative decrease $\Delta\Phi/\Phi = (\Phi_{\text{old}} - \Phi_{\text{new}})/\Phi_{\text{old}}$ is computed. The iteration stops when

- $\Delta\Phi/\Phi <$ `rtol` (10^{-4}) on `n_consec` $= 3$ consecutive steps (converged: the objective no longer decreases appreciably); or
- the number of steps reaches `max_steps` (100); or
- no trial step lowers Φ.

Requiring several consecutive small decreases avoids stopping on one accidentally small step. The log records which of the three ended the iteration, together with Φ/Φ_0, the RMS of $|\mathbf{R}|\bar{H}$, the number of CG iterations, α and the number of capped components of every step.

A small relative decrease does not by itself prove that a minimum has been reached. As a diagnostic, one further undamped Gauss–Newton step is computed at the returned state (but not taken), and its predicted relative decrease of Φ, the **Gauss–Newton decrement** $-\delta u^{\top}\mathsf{J}^{\top}\mathsf{W}\mathbf{R}/\Phi$, is reported together with the gradient norm $\|\mathsf{J}^{\top}\mathsf{W}\mathbf{R}\|$ and the stop reason. A small decrement means that the linear model sees little left to gain.

11. Above z_{cut}

Above the solved region each column is integrated hydrostatically upward from the solved pressure at level k_c, with the trapezoidal rule on the grid levels:

$$u_k - u_{k+1} + \frac{g\,(z_k - z_{k+1})}{2}\big(q(T_k,u_k) + q(T_{k+1},u_{k+1})\big) = 0, \qquad k = k_c - 1, \ldots, 1. \tag{24}$$

The equation is implicit in u_k and is solved by Newton's method (at most 30 iterations, until $|\Delta u_k| < 10^{-12}$). The result is a column-hydrostatic upper atmosphere whose pressure joins the solved one continuously at z_{cut}. If this continuation leaves the equation-of-state table anywhere, the result is marked invalid.

12. Optical depth

With P known everywhere, the table gives ρ and the 500 nm extinction χ_{500}. The vertical optical depth

$$\tau_{500}(z) = \tau_{\text{top}} + \int_z^{z_1} \chi_{500}\, dz' \tag{25}$$

is integrated from the top with the trapezoidal rule. The gas above the grid is represented by an exponential continuation of χ_{500} with the scale height $H_\chi = (z_1 - z_2)/\ln(\chi_2/\chi_1)$ of the top two levels, $\tau_{\text{top}} = \chi_1 H_\chi$ (half a grid cell, $\tfrac{1}{2}\chi_1(z_1 - z_2)$, if χ does not decrease upward there or H_χ exceeds 5000 km). The height of $\tau_{500} = 1$ is interpolated linearly in $\log_{10}\tau$ between the bracketing levels. Since the opacity grows with density, any change of pressure moves it.

13. Validity checks

The solve is attempted only if every column of the starting state stayed inside the equation-of-state table. The returned fields are checked at the end: every state inside the table, all values finite, P, ρ and χ_{500} positive, and τ_{500} strictly increasing downward in every column. The first failed check, if any, is reported with the number of points or columns concerned.

14. Three dimensions: what the result means

- Below z_{cut} the pressure balances, as well as a pressure change can, the horizontal and vertical forces of gravity and the given Lorentz force: mostly their gradient part, and through the density part of the non-gradient part. It is a minimum of Φ, not an exact equilibrium. The remaining imbalance, the buoyancy of the given temperature structure that the given field cannot hold up, would require the field (or the temperature) to change.
- Above z_{cut} the atmosphere is column-hydrostatic, not magnetohydrostatic.
- The pressure level rests on P_{ref} at the bottom of the starting state and on the anchor penalty on the mean of the bottom plane.

15. Summary of the algorithm

1. Read T, $\mathbf{B}$ and the grid; compute the starting state: column hydrostatic equilibrium with $P = P_{\mathrm{ref}}$ at the bottom of every column.
2. Compute $\mathbf{J}$ and $\mathbf{L} = \mathbf{J} \times \mathbf{B}$ once.
3. On the levels $z \le z_{\mathrm{cut}}$: compute $\bar{H}_k$ and the weights; evaluate $\mathbf{R}$ and Φ.
4. Repeat: build the layer-median preconditioner; solve the normal equations by preconditioned CG; clamp the step to $|\delta u| \le 2$; backtrack until Φ decreases; accept. Stop after three consecutive steps with $\Delta\Phi/\Phi < 10^{-4}$, after 100 steps, or when no step lowers Φ; report the Gauss–Newton decrement at the result.
5. Integrate each column hydrostatically above z_{cut}.
6. Compute ρ, τ_{500} and $z(\tau_{500} = 1)$; check the fields; write the result.